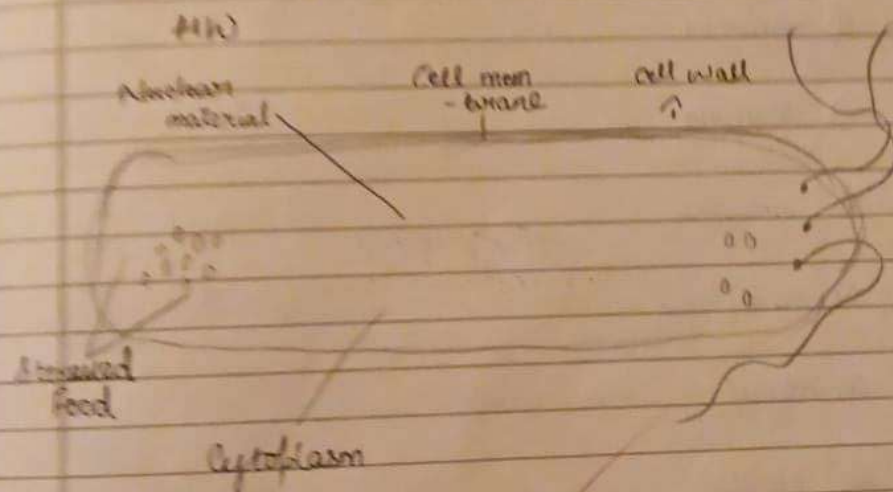


Classification of plants

K	N	L
I know about plants which are divided into different types	I want to know more about plants	I learnt what is classification I also learnt what are the 5 types of Kingdom



Generalised diagram of bacteria

Keywords :

- 1) Monera
- 2) Protista
- 3) Fungi
- 4) Plantae
- 5) Animalia
- 6) Bacteria
- 7) Amoeba
- 8) saprophytes
- 9) Algae
- 10) Mosses
- 11) Ferns
12. Angiosperm
13. ~~the~~ Gymnosperm.
14. Cryptogams
15. Phanerogams.

Q Answer the following

Q1 Classification - It is the system of arranging living organisms into groups based on their similarities and differences.

Q2 Symbiosis A relationship or interaction between two different organisms that live in harmony, each benefiting from the relationship.

Q3 Farmers often grow pea plants to naturally enrich the soil with nitrogen. Name the bacteria responsible for this and explain the process by which they make nitrogen available to plants.

Ans Rhizobium bacteria - These bacteria trap nitrogen from the atmosphere and convert it into nitrates, which can be easily absorbed by the plants along the water.

Revision worksheet - VA 1

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DATE 00 / 7 / 20

Section A

1. Choose the correct answer:

- 1) a) Protection and covering of body surfaces.
- 2) c) Fluid connective tissue
- 3) b) Both a and R are true and R is the correct explanation of A.
- 4) c) Skeletal muscle
- 5) a) Both a and R are true R is the correct explanation of A

Q. 5) Question - 2

1. a) Xylem
- b) connective tissue
- c) cuboidal epithelial tissue

- iii
1. lymph
 2. smooth muscle

lymph cell

A - 4

B - 2

C - 5

Question - 3

i Lining of the elementary canal

ii Cuticle is there above to ~~water~~ reduce water loss through transpiration and helps conserve water

Seen

21/7

iii) Answer the following:

Q1. What is Pseudopodia?

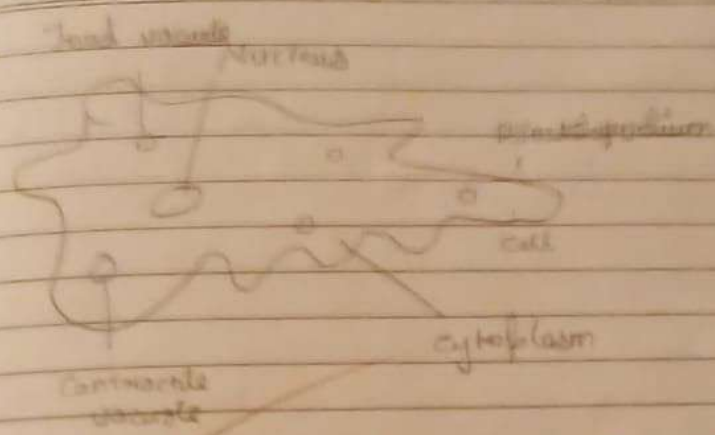
Ans It is a finger like projection formed temporarily by the extension of cytoplasm and cell membrane of amoeba along with the cell membrane in a particular direction.

Q2. Amoeba does not have any regular shape. Give reason.

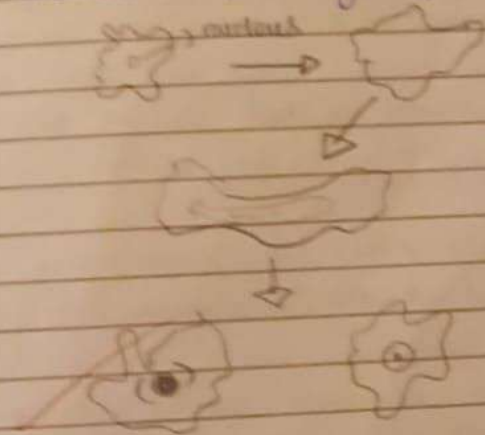
Ans Amoeba does not have cell wall, which allows them to move freely. They contain specialized structures called pseudopodia, which are bulges of the cytoplasm, given the amoeba its regular shape.

Q3. What is binary fission?

Ans A sexual reproduction in which first the nucleus divides into two, followed by the rest of the cell, ensuring each half receives one daughter nucleus.



Amoeba reproducing by binary fission



Daughter Amoeba

6. Why does amoeba undergo multiple fission only inside the cyst and not in its free living state?

Ans During unfavourable environmental conditions, an amoeba withdraws its pseudopodia and turns into a small rounded mass. The amoeba then secretes a thick wall around itself, forming a cyst. Reproduction occurs within the cyst through multiple fission.

7. A student claims that bacteria are the greatest enemy to human health. Justify this claim by discussing their roles in both causing diseases and affecting the safety of the food we eat.

Ans Bacteria are responsible for a number of diseases such as typhoid, leprosy, tuberculosis (TB), pneumonia, cholera, etc. They also cause the spoilage of food items such as milk, meat, fish and vegetables. Additionally, tinned and packaged food can spoil due to bacterial contamination after the expiry date.

12/10

ASSIGNMENT - 5

Answer the question given below

- i) Name the kingdom to which bacterium belongs:

Ans. The kingdom to which the bacterium belongs is monera.

- ii) Write two characteristic features of kingdom to which the bacterium belongs.

Ans. They are single-celled organisms that lack defined nucleus and membrane-bound organelles.

- Most organisms in this kingdom possess a rigid cell wall.

- iii) Name the gelatinous, protect covering that surrounds the cell wall of the bacterium.

Ans. Capsule

iv) Mention two cellular organelles that are absent in the bacterium.

ans ~~Nucleus~~ ^{mitochondria}, ~~Nucleolus~~, and plastids are absent.

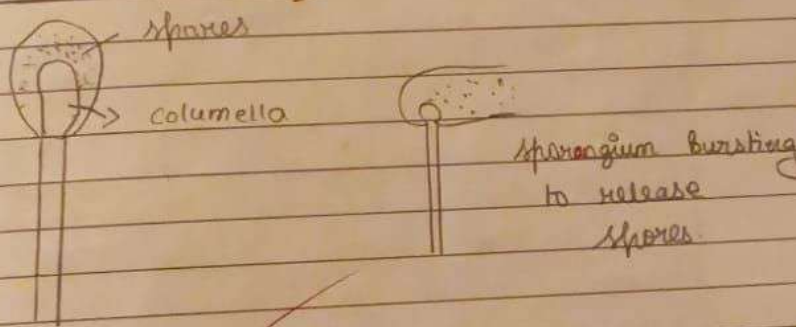
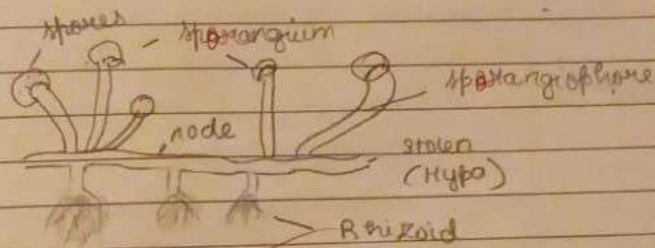
v) Write specific name of cocci shaped bacterium.

ans ~~Staphylococcus aureus~~

4.8.26



mycelium
Bread mould growing
on bread



4/6

7 What is mycelium?

Ans A network of hyphae found in bread mould.

8 Mosses grow abundantly in damp, shady places. Explain how their structure is adapted to this habitat

Ans Mosses are very small plants with tiny stems and leaf like parts but no true roots. Instead, they have thin, thread like rhizoids that help them stick to moist soil, rocks or walls and take in water.

Assignment - 6

4 Study the figure given along side and answer the following questions.

i Identify the organism. write its common name as well as the scientific name.

Ans Bread mould
Scientific name - *Rhizopus*

ii label the guidelines 1-5

- 1) Spores
- 2) sporangium
- 3) sporangiophore
- 4) stolon
- 5) Rhizoids

iii which kingdom does it belong to

Ans Fungi

iv write one important role of a part shown as 2

Ans The Spongia produces spores, opening when to release them for dispersal by reproduction.

5 Mention any very specific features of this category of plants.

Ans They are heterotrophic / saprophytic and lack chlorophyll, which they cannot perform photosynthesis and absorb nutrients from dead organic matter.

9) Why does a mango seed always found enclosed within a fleshy covering, while a pine seed lies exposed on woody scale what does this reveal about the fundamental structural difference in the reproductive design of these two plant groups?

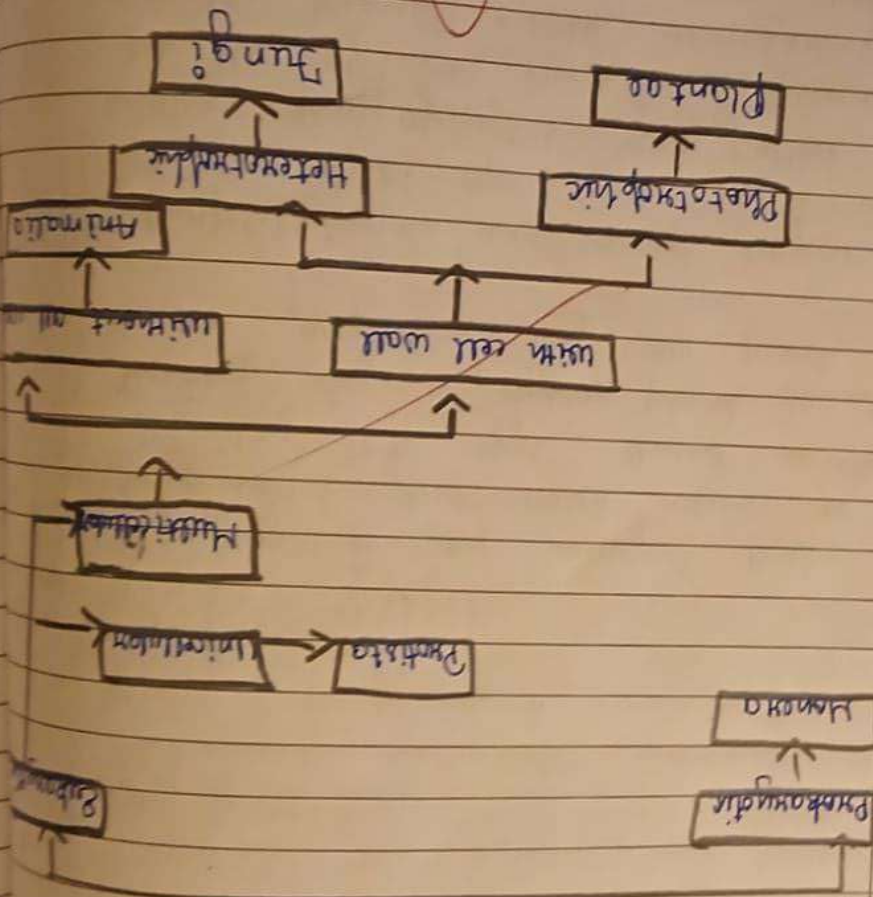
Any amphibians are these organisms which live on both land and in water. Bryophytes are called amphibian on the plant kingdom because these plants though live in soil but they need water for sexual reproduction.

10. A frog is an animal that lives on land but must return to water to lay its eggs. Based on this, justify why a moss plant is considered as amphibians of the plant world.

Ans. In angiosperms, the seed develops inside an ovary of the flower, after fertilisation. This ovary grows into a fruit. As the seeds are completely covered and protected by the fruit wall. In gymnosperms, there is no fruit on the ovary. The seed develops on the surface of some scales and remains exposed.

Concept Map

Organisms



Assignment - 7

Case study based question:

During a school trip to a bird station, students of class 7 walked past tall trees with needle-like leaves. They collected dry cones scattered beneath these trees. When opening gently, they noticed that the seeds slipped out. He noticed that the seeds were not enclosed in fruits but lay exposed on the scales of the cones.

1) Do these trees belong to flowering plants? If not, which group do they belong to?

Ans No, these trees do not belong to flowering plants. They belong to a group of non-flower plants called gymnosperm.

2) Give the examples of plants that belong to the same category.

Photosynthesis and Respiration

35

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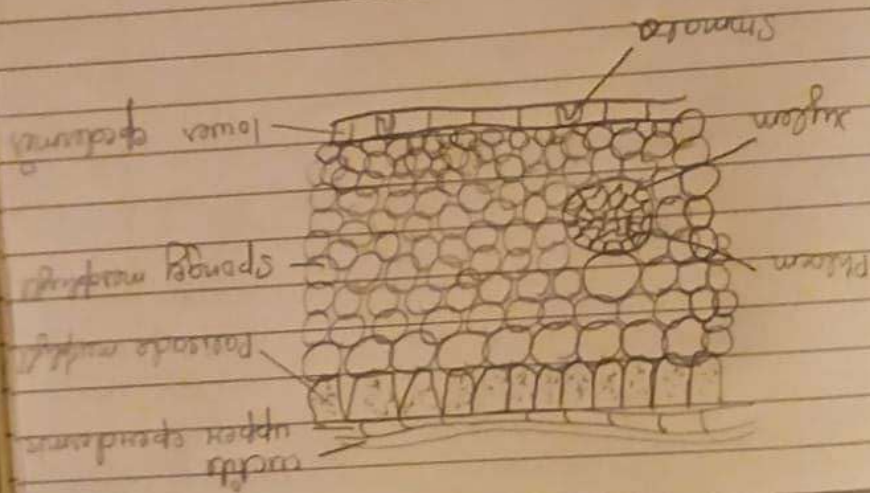
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1

I want to know about photosynthesis, autotrophic, nutrition, vascular system.

(1)

K



1) Keywords :

1. Autotrophs
2. photosynthesis
3. chloroplasts
4. palisade mesophyll
5. chlorophyll
6. stomata
7. guard cells
8. variegated leaves
9. hydra
10. phloem
11. xylem
12. iodine
13. alcohol
14. starch
15. glucose.

II Answer the following :

1 Explain the following terms :

- a) Autotrophs - Organisms including green plants, algae, and some bacteria, that prepare their own food through process called photosynthesis.
- b) Photosynthesis - The process by which green plants, algae, and certain bacteria synthesise glucose from carbon dioxide and water, in the presence of sunlight and chlorophyll, releasing oxygen as a by-product.
- c) Vascular system - The transport system of plants is composed of two types of conducting tissues xylem and phloem. Xylem transports water and transported mineral from the roots to all parts of the plant. Phloem transports prepared food (sucrose) from the leaves to all other parts of the plant. Both upward & downward.